

Yechan Pyeong

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Education

Korea Advanced Institute of Science and Technology (KAIST), Feb 2022 – Feb 2028
Daejeon, Republic of Korea (expected)

- B.S. in Chemistry (Advanced Major)
- Cumulative GPA: 4.16/4.3 (98.60/100)

Chung-nam Science High School, Gongju, Republic of Korea Mar 2020 – Jan 2022

Research Interests

- Theoretical and Computational Chemistry
- Chemical Reaction Dynamics and Nonadiabatic Processes
- Exploration of Potential Energy Surfaces (PES)
- Development of Numerical Algorithms for Molecular Systems

Research Experience

Semiclassical nonadiabatic dynamics and electronic coherence in photochemical reactions (Advisor: Prof. Young Min Rhee) Dec 2025 - Present

- Analyzed mixed quantum-classical (pbme-nH) trajectories to determine the role of electronic coherence in furan photochemistry.
- Investigated how phase-coherent electronic dynamics influence branching ratios at conical intersections during ring-opening reactions.
- Integrating Spin Mapping into the pbme-nH framework to replace MMST mapping, aiming to improve the physical accuracy of semiclassical simulations.
- Studied advanced theoretical frameworks, including AIMS, FSSH, and Basin-hopping algorithms, for potential energy surface exploration.

Automated bond perception and atom mapping via mixed-integer linear programming (Advisor: Prof. Woo Youn Kim) Jun 2023 – Nov 2023

- Developed core modules for the *acerxn* program to predict reaction networks and determine bond orders from 3D atomic connectivity.
- Formulated the Atom Connection to Bond Order (ACtoBO) problem as a MILP model to restore bond orders and formal charges from Cartesian coordinates.
- Optimized multiple objective functions with Gurobi and SciPy, successfully resolving resonance and charge polarization errors.
- Achieved >99% bond order restoration accuracy by benchmarking against thousands of complex organic molecules from PubChem.

Relevant Coursework

Theoretical & Computational Chemistry

- CH40011: Introduction to Molecular Spectroscopy (A+)
- CH40012: Chemical Reaction Dynamics (A+)
- CH40015: AI Chemistry (In Progress, Fall 2026)
- CH50012: Quantum Chemistry I (In Progress, Fall 2026)

Advanced Chemistry Foundations

- CH30011: Physical Chemistry III (A+)
- CH30022: Physical Organic Chemistry (A+)

Mathematics & Programming

- MAS10009: Introduction to Linear Algebra (A+)
- MAS20050: Probability and Statistics (A+)

Leadership & Extracurricular Experience

Principal Double Bassist, KAIST Orchestra, Mar 2023 - Present
Daejeon, Republic of Korea

- Perform as a dedicated double bassist for regular symphonic concerts and serve as the Section Leader (from Jan 2026) to direct rehearsals and mentor section members.

Prescribed Load List (PLL) Specialist / KATUSA, Jan 2024 – Jul 2025
Eighth U.S. Army, Republic of Korea

- Managed logistical support for a U.S. Army combat regiment while facilitating cross-cultural communication and operational coordination during major US-ROK joint military exercises.

Active Member, RUN (KAIST Algorithm Club), Mar 2022 – Dec 2023
Daejeon, Republic of Korea

- Strengthened coding proficiency through algorithm seminars and programming competitions.

Scholarship & Fellowship

National Science and Engineering Scholarship, Spring 2026 - Present
Korea Student Aid Foundation (KOSAF)

KAIST Alumni Academic Scholarship, Spring 2026 - Present
KAIST Alumni Association

Academic Excellence Scholarship (Top of the Class), Fall 2025
Department of Chemistry, KAIST

Dean's List, Department of Chemistry, KAIST Fall 2025

Dean's List, School of Freshman, KAIST Spring 2022

Skills

Computational Chemistry Tools: BAGEL, GROMACS, Q-Chem, GAFF

Programming Languages: C, C++, Python, MATLAB

Languages: English (Fluent), Korean (Native)

References

Prof. Young Min Rhee (ymrhee@kaist.ac.kr) KAIST

<https://singlet.kaist.ac.kr/>

Prof. Woo Youn Kim (wooyoun@kaist.ac.kr) KAIST

<https://wooyoun.kaist.ac.kr/>